Using Concepts of Wheatstone Bridges to Measure Weight

ME104

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May 15, 2025

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Objective

Accurate measurement of mass is essential in engineering applications or everyday contexts. In professional facilities, mass balances cost upwards hundreds of thousands of dollars, rendering them unobtainable for many small-scale uses. This project demonstrates that the use of low-cost electronics is viable to produce an accurate scale. These are the given requirements and restraints:

1. Spend less than $30 on materials, outside of Arduino MEGAs, HX711s, jumpers, buttons, and LEDs.
2. The scale must have two button functions: one to TARE the weight and one to switch between units of measurements.
3. Define the following criteria:
    a. Maximum weight
    b. Percent error
    c. Linearity
    d. Consistency
4. Have an easy-to-use interface.

Design Analysis

After considering the different objectives of the project, the team decided on a Wheatstone bridge configuration to measure weight. The team chose to have four parallel strain gauges to maximize sensitivity in detecting the bending of the metal, which has a high Young's modulus. By putting two strain gauges on top and two on the bottom, both the compression and tension of the metal were able to be considered when calculating the mass of the object on the scale. Below is the analysis of the electronic, mechanical and algorithmic subsystems/components of the solution.

Wheatstone Bridge and Amplifier

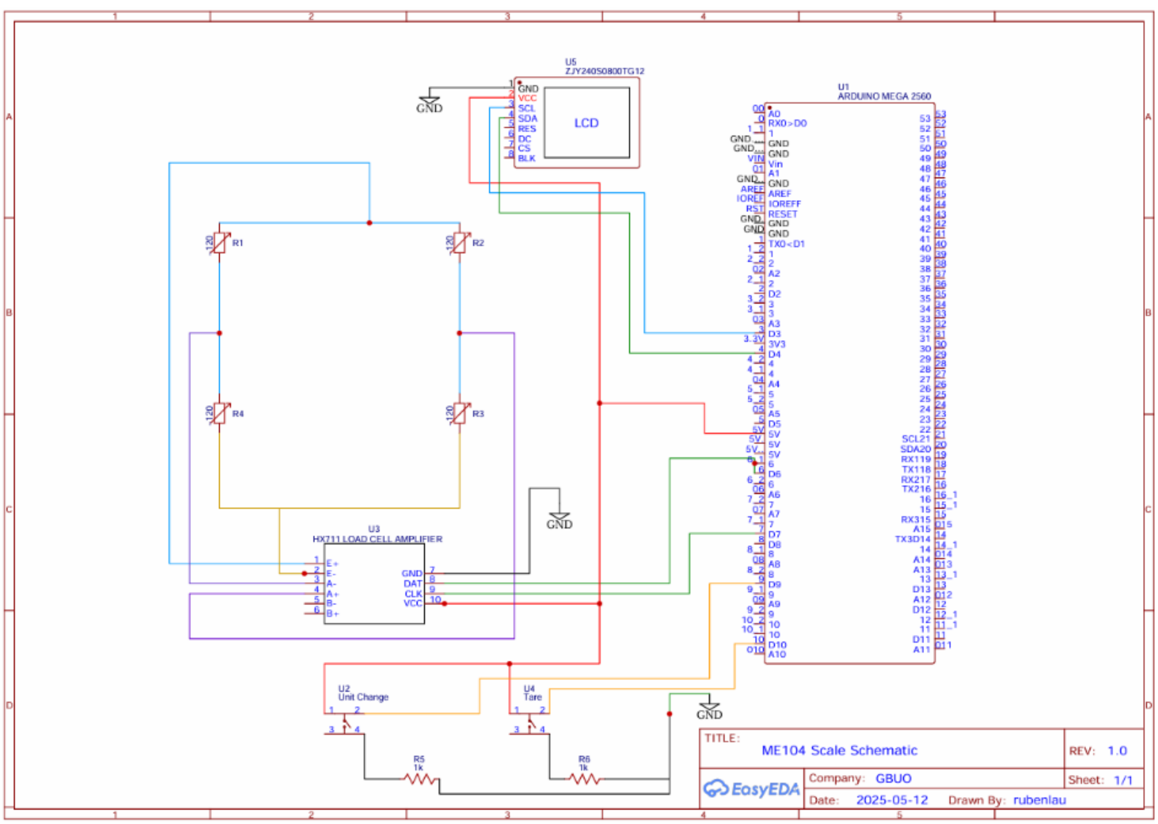

Figure 1. Electronics Schematic

The main electronic component of the system functions by arranging the four different strain gauges in the orientation of a Wheatstone bridge, allowing a voltage difference to be measured across A- and A+, and therefore the strain of the entire platform. The orientation and position of the strain gauges is crucial. The strain gauge must be oriented parallel to the long side of the platform to properly read resistance changes when put in tension or compression. Further, strain gauges R1 and R3 must be on one side of the platform, while the R2 and R4 must be on the other, shown in the diagram below:

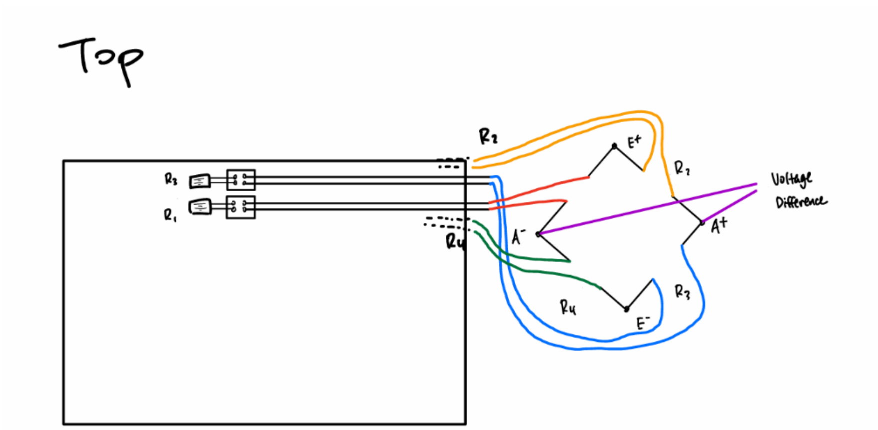

Figure 2. Strain Gauge Orientation and Wheatstone Bridge Diagram

This is because R1, R4 and R2, R3 are on their respective sides of the bridge; if they were put on the same side, the measured difference in voltage would not change (assuming ideal conditions). Consequently, on the final design, R1, R3 and R2, R4 were put on the same sides. This voltage differential across A+ and A- was converted and amplified by the 24-bit ADC in the HX711 into bit values that the Arduino could measure and read. The bit measurements would be calibrated by the code into the final measurement of weight in respective units, explained later.

The team encountered the issue of noise and fluctuation of the bit measurements amplified by the HX711 board, shown in the schematic above. By testing the measurements with different modes of connectivity, the team found that the breadboard connection between the bridge and the HX711 caused much of the unstable output, which did not allow a unique reading to be taken. Figure 3 displays the original wiring of the system to the breadboard, featuring solderless connections.

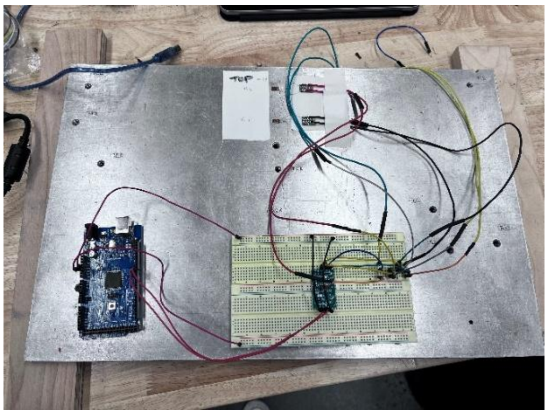

Figure 3. Initial Assembly

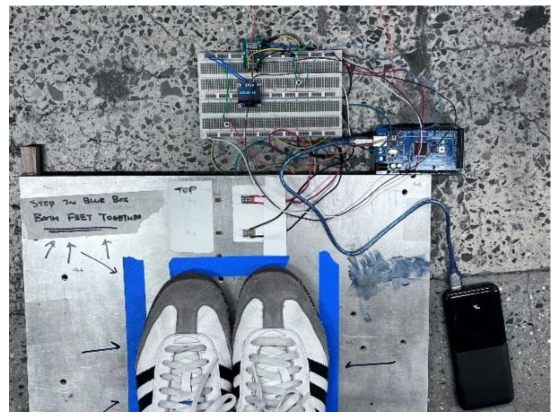

Figure 4. Final Assembly, 1

Therefore, the team had to iterate by soldering the connections of the strain gauges directly to the HX711 and using brand new breadboards to decrease the factor of noise on the amplifier (note that Figure 4 shows the HX711 hanging off the breadboard, allowing the wires to be soldered to E+, E-, A+ and A- pins). Troubleshooting was much of the iterative process, such as finding ineffective connection points with multimeters.

LCD

The second electronic component of the system is to create a visual read-out, presented on an LCD, which was supplied to the group. The LCD was wired as shown in Figure 1. The LCD allowed the group to use a portable power source, more convenient in comparison to reading values off the serial monitor on Arduino IDE.

Buttons

The user interacts with the scale via two buttons: one for tare and the other for switching between units. Each button was directly wired to the 5V rail, its input pin on the Arduino, and a 10k Ohm pull down resistor connected to ground. This ensures that the button stays on low when the button is not being pressed. When the button is pressed, it switches to high, which indicates to the Arduino that it should either switch between units, or tare.

Platform

The main mechanical component of the system is the platform. The platform consists of an aluminum board with two wooden rectangles that act as feet attached along the width. Note the two protruding wooden pieces in Figure 5 below.

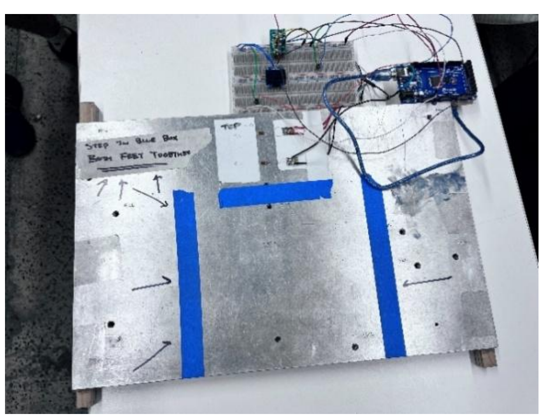

Figure 5. Final Assembly, 2

The wooden feet are essential to allowing the ¾" thick steel board bend under stress as a simply supported cantilever, where the load is applied in the middle with supports at the ends.

A challenge with this configuration arises due to the strain gauges being situated at the top of the wide board, making the bending inconsistent along the width of the board. Therefore, there is a large effect of where the load is placed on the measurement of strain. The blue taped off area displays the general location of which the load should be placed to give accurate readings.

Code

```
#include "HX711.h"
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#define SCREEN_WIDTH 128 // OLED display width, in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels

// Declaration for an SSD1306 display connected to I2C (SDA, SCL pins)
#define OLED_RESET     -1 // Reset pin # (or -1 if sharing Arduino reset pin)
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

HX711 scale;

uint8_t dataPin = 6;
uint8_t clockPin = 7;
uint8_t tarePin = A0;
uint8_t unitPin = A1;
long calibration= -8200;

bool useKg = true;

float f;
//  adjust pins if needed



void setup()
{
  Serial.begin(115200);
  pinMode(tarePin,INPUT);
  pinMode(unitPin,INPUT);
   if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("SSD1306 allocation failed"));
    for (;;) ;       // stop here if display isn't found
  }
    display.clearDisplay();
  display.setTextSize(2);           // 2× font
  display.setTextColor(SSD1306_WHITE);

  scale.begin(dataPin, clockPin);
  scale.set_scale(calibration);
  scale.tare(20);   //  reset the scale to zero = 0
  delay(200);
}

void loop()
{
  //  continuous scale 4x per second
  if (digitalRead(unitPin) == HIGH) {
    useKg = !useKg;
    while (digitalRead(unitPin) == HIGH);
  }

  if (digitalRead(tarePin) == HIGH) {
    scale.tare(20);
    delay(200);
    while (digitalRead(tarePin) == HIGH);
  }

  float kg = scale.get_units(5);

  float w  = useKg ? kg : kg * 2.20462;

  Serial.print(w, 3);
  Serial.println(useKg ? F(" kg") : F(" lb"));
  char buf[16];
  dtostrf(w, 6, 2, buf);           // " 63.42" etc.
  display.clearDisplay();
  display.setCursor(0, 16);
  display.print(buf);
  display.print(useKg ? F(" kg") : F(" lb"));
  display.display();

  delay(25);
}
```

Figure 6. Arduino Code

The code, after initializing, takes the average of the first twenty scale readings and sets that value as zero; it was found to be more effective than taring it with one value because the values tend to fluctuate in the beginning. The values then go to the LCD monitor, and the code refreshes every 25 milliseconds. Pressing the tare button again takes the average of 20 values and sets that as the zero value of the scale. Pressing the unit button switches the state of the pin from high to low, and alternates between units. The initial state of the scale is that it uses kilograms, and upon pressing the button, it converts to pounds by multiplying a conversion factor of 2.20462.

Data Analysis

Maximum Weight

The platform used for the scale is made of an aluminum board with a high Young's modulus where it is challenging to flex and deform. Therefore, the maximum weight of the scale is undetermined. There was no calibrated weight of high enough mass at hand, so the maximum accurate weight is not known. However, upon testing, it was accurate up to 300 lbs.

Percent Error

To determine the percent error of the scale, calibrated weights were weighed and recorded. The percent error of the scale is calculated with this equation:

$$Percent\ Error = \left|\frac{Measured\ Weight - Nominal\ Weight}{Nominal\ Weight}\right| * 100\%$$

The following table displays measurement and percent error data:

Nominal Weight (g)	Measured Weight (g)	Percent Error (%)
100	89	11.0
200	140	30.0
300	250	16.7
500	446	10.8
1000	992	0.8
2000	2079	3.95

Table 1. Nominal Weights vs. Measured Weights, Percent Error

The mean percent error, derived by adding all respective values and dividing by the number of entries, is approximately 12.2%. This value is not ideal, but it also does not reflect the true accuracy of the scale. The platform picked was intended to be used to measure body weight, which is much heavier than the calibrated weights used to perform an analysis on.

Linearity

Linearity can be easily determined and visualized using a graph. Using Table 1 again, points can be plotted on a nominal weight vs. measured weight graph.

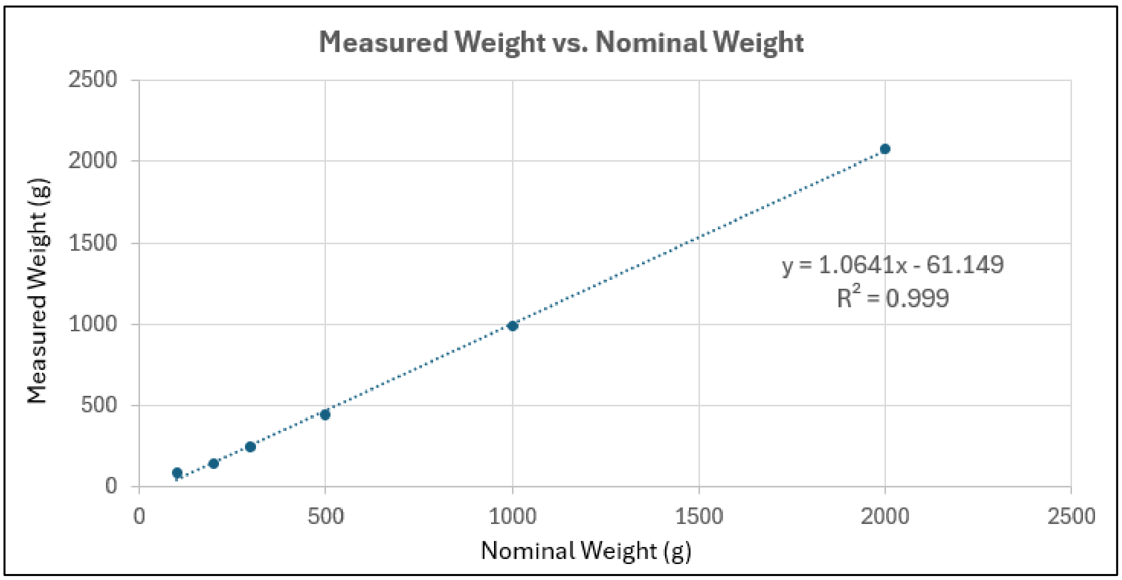

Figure 7. Nominal Weights vs. Measured Weights, Percent Error

The R-squared value is a statistical measure that indicates how well a data set adheres to a linear regression model. If R-squared value is closer to 1, the more linear the graph is and vice versa. This graph displays an R-squared value of 0.999, which suggests that the scale effectively maintains the same error throughout the given range of weights. This is desired – proportional readings are given for different weights. This also suggests that there is just a systematic error with our device, rather than random errors. Consequently, the scale can likely be further improved through calibration rather than a change in the entire system.

Consistency

To quantify the consistency of the scale, a constant nominal 2000g weight was measured thirty times. The scale was not tared between measurements as that would defeat the purpose of measuring the consistency.

Trial	Weight (kg)	Trial	Weight (kg)	Trial	Weight (kg)
1	2.09	11	2.13	21	2.21
2	2.22	12	2.12	22	2.14
3	2.10	13	2.12	23	2.17
4	2.16	14	2.11	24	2.14
5	2.11	15	2.02	25	2.19
6	2.13	16	2.21	26	2.22
7	2.16	17	2.15	27	2.16
8	2.12	18	2.16	28	2.18
9	2.14	19	2.18	29	2.19
10	2.22	20	2.19	30	2.13

Table 2. Weight Measurements for Thirty Trials

Using this data, consistency can be quantified by calculating the average, standard deviation, and coefficient of variation. The following equations were used for the respective quantities:

$$Mean = \bar{x} = \frac{1}{n} * \sum_{i=1}^{n} x_i$$

$$Standard\ Deviation = \sigma = \sqrt{\frac{1}{n} * \sum (x_i - \bar{x})^2}$$

$$Coefficient\ of\ Variation = \frac{\sigma}{\bar{x}} * 100\%$$

The average is calculated to be approximately 2.15. This value represents the overall measurement of the weights across all trials. This is a simple average.

The standard deviation is calculated to be approximately 0.043. Standard deviation measures the spread of the scale readings. A lower value indicates greater precision. The calculated value is low, indicating little variation and high consistency.

The coefficient of variation is calculated to be approximately 2.02%. The coefficient of variation is a measure of variability, determining the consistency of measurements compared to the mean value. A lower value indicates greater precision. The calculated percentage is low, indicating high reliability and consistency.

Conclusion

All objectives were met; less than $30 was spent on the materials, the scale had a dedicated tare and units button, and an LED screen displayed the current readings on the scale. The scale also displays linear error, where the R^2 value was calculated to be 0.999 between the nominal weight and the measured weight, meaning the scale achieved measurements very close to being 1:1 with the nominal weight. With a standard deviation of 0.043kg through 30 readings, the scale is very self-consistent. A linear error indicates that with further calibrations, the error could be further decreased, as it is a systematic error rather than a random error.

By sacrificing temperature sensitivity, the scale increased sensitivity on readings. Moving on, it could be possible to test for temperature sensitivity and reduce reading sensitivity. It is also possible that the problem stemmed from the use of strain gauges, which give different readings depending on where the object is placed, so it may be in the best interest of the group to use load cells for more accurate readings.